

Linear First-Order Equations & the Integrating Factor

ANSWER KEY



Recognizing linear form

- 1 Write $y' + 2xy = x$ in the standard form $y' + P(x)y = Q(x)$ and identify $P(x)$ and $Q(x)$.

Already in standard form: $P(x) = 2x$, $Q(x) = x$.

- 2 Is $y' = \frac{y}{x} + x^2$ linear? Rewrite it in standard form.

Yes. Standard form: $y' - \frac{1}{x}y = x^2$, so $P(x) = -\frac{1}{x}$, $Q(x) = x^2$.

Finding the integrating factor

- 3 Find the integrating factor for $y' + 3y = e^x$.

$$\mu(x) = e^{\int 3 dx} = e^{3x}.$$

- 4 Find the integrating factor for $y' - \frac{2}{x}y = x$, $x > 0$.

$$\mu(x) = e^{\int -2/x dx} = e^{-2 \ln x} = x^{-2}.$$

Solving with the integrating factor

- 5 Solve $y' + 3y = e^x$.

With $\mu = e^{3x}$: $(e^{3x}y)' = e^{4x}$, so $e^{3x}y = \frac{1}{4}e^{4x} + C$, giving $y = \frac{1}{4}e^x + Ce^{-3x}$.

- 6 Solve $y' - \frac{2}{x}y = x$, $x > 0$.

With $\mu = x^{-2}$: $(x^{-2}y)' = x^{-1}$, so $x^{-2}y = \ln x + C$, giving $y = x^2 \ln x + Cx^2$.

Initial value problems

- 7 Solve $y' + y = 2$, $y(0) = 0$.

With $\mu = e^x$: $(e^x y)' = 2e^x$, so $e^x y = 2e^x + C$, giving $y = 2 + Ce^{-x}$. Since $y(0) = 0$, $C = -2$: $y = 2 - 2e^{-x}$.

- 8 Solve $y' + 2y = 4x$, $y(0) = 1$.

With $\mu = e^{2x}$: $(e^{2x}y)' = 4xe^{2x}$. By parts, $\int 4xe^{2x} dx = 2xe^{2x} - e^{2x} + C$. So $y = 2x - 1 + Ce^{-2x}$. Since $y(0) = 1$, $C = 2$: $y = 2x - 1 + 2e^{-2x}$.

Application: a filling tank

- 9** A tank starts with 100 L of pure water. Brine with concentration 2 kg/L flows in at 3 L/min, and the well-mixed solution flows out at the same rate. If $Q(t)$ is the salt (kg) at time t , the balance gives $Q' = 6 - 0.03Q$. Solve for $Q(t)$ given $Q(0) = 0$.
- Standard form $Q' + 0.03Q = 6$. With $\mu = e^{0.03t}$: $(e^{0.03t}Q)' = 6e^{0.03t}$, so $e^{0.03t}Q = 200e^{0.03t} + C$, giving $Q = 200 + Ce^{-0.03t}$. Since $Q(0) = 0$, $C = -200$: $Q(t) = 200(1 - e^{-0.03t})$.
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More integrating factor practice

- 10** Find the integrating factor for $y' + \frac{1}{x}y = \cos x$, $x > 0$.

$$\mu(x) = e^{\int (1/x) dx} = e^{\ln x} = x.$$

- 11** Solve $y' + \frac{1}{x}y = \cos x$, $x > 0$.

With $\mu = x$: $(xy)' = x \cos x$. By parts, $\int x \cos x dx = x \sin x + \cos x + C$. So $y = \sin x + \frac{\cos x + C}{x}$.

More initial value problems

- 12** Solve $y' - 2y = 4$, $y(0) = 3$.

$\mu = e^{-2x}$: $(e^{-2x}y)' = 4e^{-2x}$, so $e^{-2x}y = -2e^{-2x} + C$, giving $y = -2 + Ce^{2x}$. Since $y(0) = 3$, $C = 5$: $y = -2 + 5e^{2x}$.

- 13** Solve $y' + \frac{y}{x} = 3x^2$, $y(1) = 2$, $x > 0$.

$\mu = x$: $(xy)' = 3x^3$, so $xy = \frac{3}{4}x^4 + C$, giving $y = \frac{3}{4}x^3 + \frac{C}{x}$. Since $y(1) = 2$, $C = \frac{5}{4}$: $y = \frac{3}{4}x^3 + \frac{5}{4x}$.

Application: an RL circuit

- 14** An RL circuit satisfies $L \frac{di}{dt} + Ri = V$. With $L = 2$, $R = 10$, $V = 20$, and $i(0) = 0$, solve for $i(t)$.

$2i' + 10i = 20 \Rightarrow i' + 5i = 10$. With $\mu = e^{5t}$: $(e^{5t}i)' = 10e^{5t}$, so $e^{5t}i = 2e^{5t} + C$, giving $i = 2 + Ce^{-5t}$. Since $i(0) = 0$, $C = -2$: $i(t) = 2(1 - e^{-5t})$.

A second mixing-tank variant

- 15** A 200 L tank starts with 50 kg of dissolved salt. Pure water flows in at 5 L/min and the mixture flows out at the same rate. Find $Q(t)$, the salt (kg) at time t .

Balance: $Q' = 0 - \frac{Q}{200}(5) = -0.025Q$. With $\mu = e^{0.025t}$: $(e^{0.025t}Q)' = 0$, so $Q = Ce^{-0.025t}$. Since $Q(0) = 50$:

$$Q(t) = 50e^{-0.025t}.$$

Checking a solution by substitution

- 16** Verify that $y = 3 - 2e^{-x}$ solves $y' + y = 3$.

$y' = 2e^{-x}$. Then $y' + y = 2e^{-x} + 3 - 2e^{-x} = 3$, matching the equation.

One more round of practice

- 17** Solve $y' + 4y = 8$, $y(0) = 5$.

With $\mu = e^{4x}$: $(e^{4x}y)' = 8e^{4x}$, so $e^{4x}y = 2e^{4x} + C$, giving $y = 2 + Ce^{-4x}$. Since $y(0) = 5$, $C = 3$:
 $y = 2 + 3e^{-4x}$.

- 18** Find the integrating factor for $y' + \frac{2}{x}y = 3x$, $x > 0$, and solve.

$\mu(x) = e^{\int (2/x) dx} = x^2$. Then $(x^2y)' = 3x^3$, so $x^2y = \frac{3}{4}x^4 + C$, giving $y = \frac{3}{4}x^2 + \frac{C}{x^2}$.

One final integrating-factor problem

- 19** Solve $y' - 3y = 6$, $y(0) = 1$.

With $\mu = e^{-3x}$: $(e^{-3x}y)' = 6e^{-3x}$, so $e^{-3x}y = -2e^{-3x} + C$, giving $y = -2 + Ce^{3x}$. Since $y(0) = 1$, $C = 3$:
 $y = -2 + 3e^{3x}$.

- 20** Find the integrating factor for $y' + 5y = e^{-2x}$.

$\mu(x) = e^{\int 5 dx} = e^{5x}$.

- 21** Solve $y' + 5y = e^{-2x}$ using the integrating factor found above.

$(e^{5x}y)' = e^{5x}e^{-2x} = e^{3x}$, so $e^{5x}y = \frac{1}{3}e^{3x} + C$, giving $y = \frac{1}{3}e^{-2x} + Ce^{-5x}$.

- 22** Solve $y' - y = e^{2x}$, $y(0) = 0$.

With $\mu = e^{-x}$: $(e^{-x}y)' = e^x$, so $e^{-x}y = e^x + C$, giving $y = e^{2x} + Ce^x$. Since $y(0) = 0$, $1 + C = 0 \Rightarrow C = -1$:
 $y = e^{2x} - e^x$.

A closing pair of mixed problems

- 23** Find the integrating factor for $y' + \frac{3}{x}y = x^2$, $x > 0$.

$\mu(x) = e^{\int (3/x) dx} = e^{3 \ln x} = x^3$.

- 24** Solve $y' + \frac{3}{x}y = x^2$ using the integrating factor found above.

$(x^3y)' = x^3 \cdot x^2 = x^5$, so $x^3y = \frac{1}{6}x^6 + C$, giving $y = \frac{1}{6}x^3 + \frac{C}{x^3}$.